

# Akansha

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## EDUCATION

### Indian Institute of Technology Kharagpur

*B.Sc., Mathematics and Computing*

Kharagpur, India

2022 – 2026 (*Expected*)

- **Relevant Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Probability & Statistics

## PROJECTS

### Connexa | *Next.js 15, TypeScript, Express.js, Prisma, WebSockets* | GitHub | Live

Jun 2025 – Present

- Engineered a real-time full-stack chat platform with one-on-one and group messaging, read receipts, presence tracking, and typing indicators using **Next.js 15**, **TypeScript**, and **WebSockets**
- Built an event-driven backend with **Express.js** and **Prisma ORM**, applying query optimization for low-latency messaging under concurrent load
- Implemented secure **JWT authentication** with rotating tokens, HTTP-only cookies, and email verification and recovery via Nodemailer and SendGrid
- Built a responsive frontend with **Zustand** and **Tailwind CSS**, using selective state subscriptions to minimize unnecessary re-renders

### SynqX | *FastAPI, Celery, Redis, React, TypeScript, React Flow* | GitHub

Dec 2025 – Present

- Created an open-source **ETL orchestration platform** to visually design, schedule, and monitor data pipelines through a drag-and-drop **DAG interface**
- Designed a distributed, event-driven system using **FastAPI**, **Celery**, and **Redis** with isolated control and data planes for scalable, fault-tolerant execution
- Implemented job scheduling and retry logic with live telemetry over **WebSockets** for real-time pipeline observability
- Built an interactive frontend in **React** and **TypeScript** using **React Flow** for real-time DAG visualization

### Windows Calculator Clone | *React, Styled Components, Nerdamer* | GitHub | Live

Feb 2024 – May 2024

- Developed a multi-mode calculator replicating Windows 11 Standard, Scientific, Programmer, and BMI modes in **React** with **React Router v6**
- Integrated **Nerdamer** for symbolic algebra and real-time converters across physical, digital, and currency units
- Designed a modular, theme-aware UI using **React Router v6** and **Styled Components** for adaptive layouts

### NASA Space Explorer | *React, Vite* | GitHub | Live

Jun 2024

- Built a **React + Vite** portal integrating NASA APIs (APOD, Mars Rover, Media Library) with async rendering and lazy-loaded components
- Built reusable hooks and components to reduce duplication and improve maintainability
- Delivered a responsive, theme-aware UI with **React Context** for consistent cross-device experience

## TECHNICAL SKILLS

**Languages:** C, C++, Python, JavaScript (ES6+), TypeScript, SQL

**Frameworks:** React, Next.js, Node.js, Express.js, FastAPI, Celery

**Databases and Tools:** PostgreSQL, MongoDB, Redis, REST APIs, Git, GitHub

**Concepts:** Data Structures, Algorithms, OOP, System Design, Microservices, CI/CD, API Design

## ACHIEVEMENTS

**JEE Advanced 2022:** Secured All India Rank **27,182**, earning admission to IIT Kharagpur (Mathematics and Computing)

**Flipkart Runway Season 4:** Shortlisted for interviews in Flipkart's national internship program for women engineers

**Wing Representative, Mother Teresa Hall:** Managed residential operations for 100+ students; led logistics, welfare, and inter-hall coordination

**Illumination 2023 (IIT Kharagpur):** Contributed to structure design and coordination for a 100+ member team in the institute's premier cultural and design competition